



ST BENEDICT'S

SUBJECT
GRADE
EXAMINER
NAME
TEACHER

Mathematics
11
Mrs K Myburgh
Memorandum

PAPER
DATE
MARKS
MODERATOR
DURATION

2
11 November 2020
150
Grade 11 teachers
3 hours

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
	SECTION A:		
1	Statistics	11	
2	Analytical geometry	14	
3	Trigonometry	18	
4	Euclidean geometry	14	
5	Measurement	9	
6	Trigonometric functions	12	
	SECTION B:		
7	Statistics	12	
8	Trigonometry	18	
9	Solving triangles	11	
10	Euclidean geometry	5	
11	Euclidean geometry	12	
12	Analytical geometry	14	
TOTAL		150	

INSTRUCTIONS:

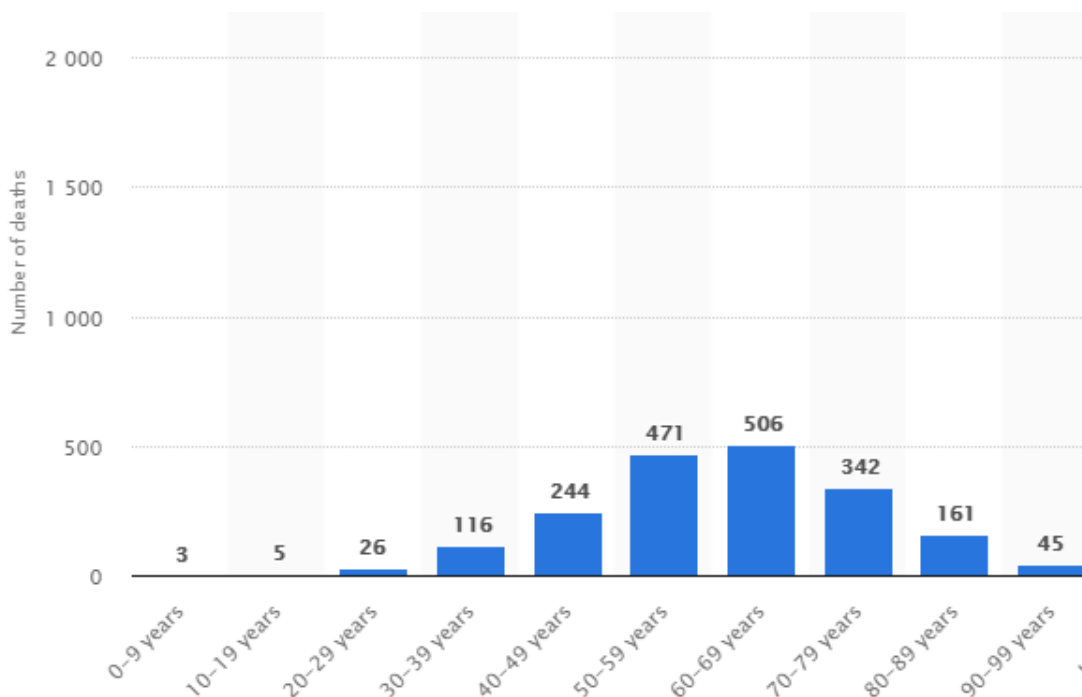
1. This paper consists of 12 questions and 27 pages. A separate information sheet has been given to you.
2. Read the questions carefully.
3. Answer all questions in the spaces provided.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Round off your answers to two decimal digits where necessary.
7. All necessary working details must be shown. Answers only, without the relevant calculations will not be given marks. Equations may not be solved solely with a calculator.
8. It is in your interest to write legibly and present your work neatly.
9. Two blank pages (pages 25 to 26) are included at the end of the paper. If you run out of space for a question, use these pages. Clearly indicate the question number of your answer should you use this extra space.

SECTION A:

QUESTION 1

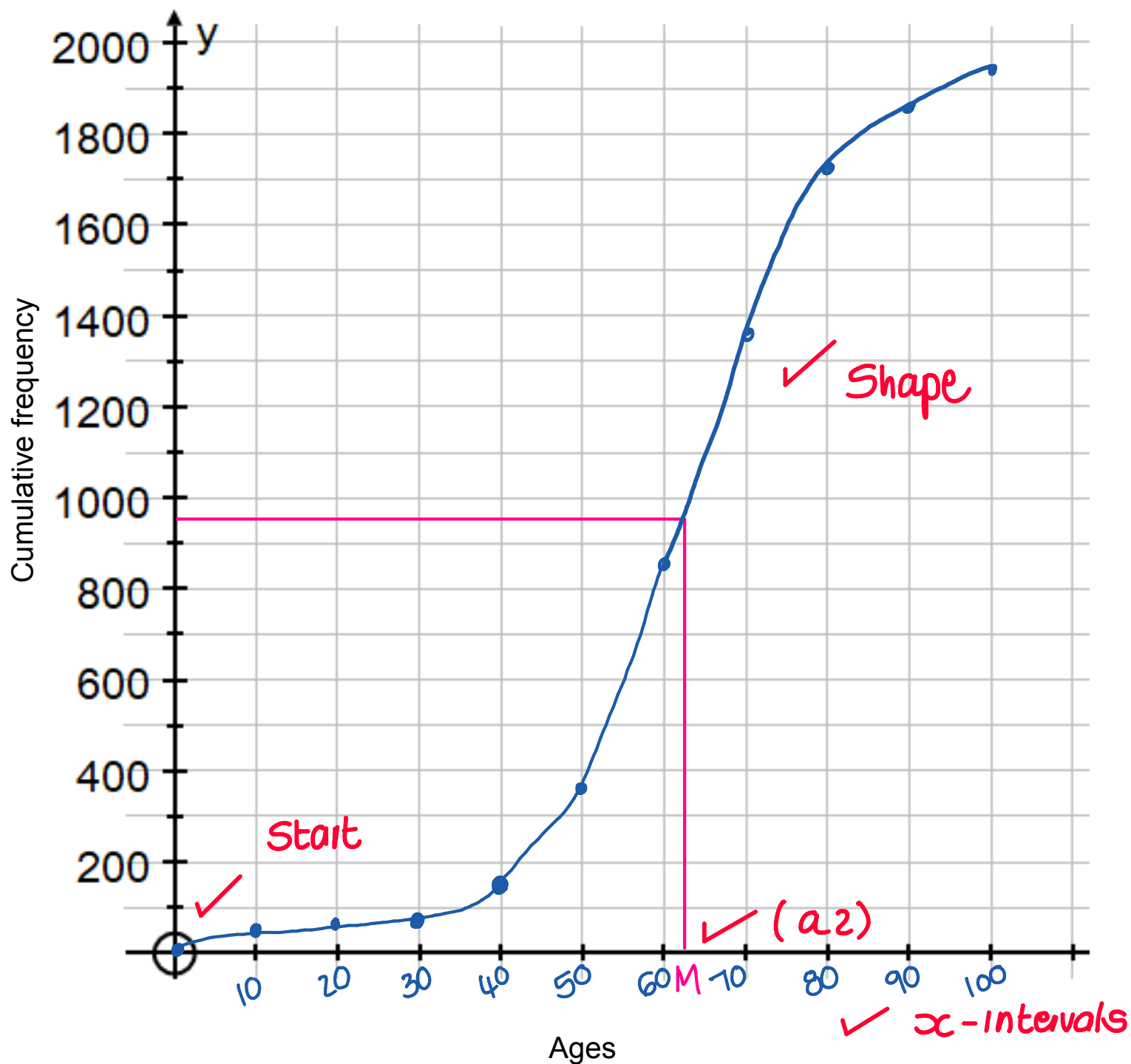
11 MARKS

The following bar graph shows the number of coronavirus (COVID-19) deaths in South Africa by 22 June 2020, by age.



- a)
- 1) Complete the following table and construct a cumulative frequency graph (ogive) on the grid provided. (5)

AGE	FREQUENCY	CUMULATIVE FREQUENCY	
$0 < x < 10$	3	3	Co-ordinates: <u>(0;0)</u> (10;3) (20;8) (30;34) (40;150) (50;394) (60;865) (70;1371) (80;1713) (90;1874) (100;1919)
$10 \leq x < 20$	5	8	
$20 \leq x < 30$	26	34	
$30 \leq x < 40$	116	150	
$40 \leq x < 50$	244	394	
$50 \leq x < 60$	471	865	
$60 \leq x < 70$	506	1371	
$70 \leq x < 80$	342	1713	
$80 \leq x < 90$	161	1874	
$90 \leq x \leq 100$	45	1919	



- 2) Use your graph to estimate the median age, showing where you take your reading with the letter M. (2)

$$Q_2 = 63 \quad \checkmark$$

- 3) Determine the estimated mean age of the data and comment on the skewness of the data. (3)

$$\bar{x} = 61,59 \checkmark$$

→ ca for skewness. ✓✓

- b) The following is an extract from President Cyril Ramaphosa's address on the 17th June 2020.

“We took a decision early in the development of the disease in South Africa ... with the declaration of a National State of Disaster ... In doing so, we aimed to ‘flatten the curve’ ...”

Making reference to your ogive from question a, explain what is means to ‘flatten the curve’. (1)

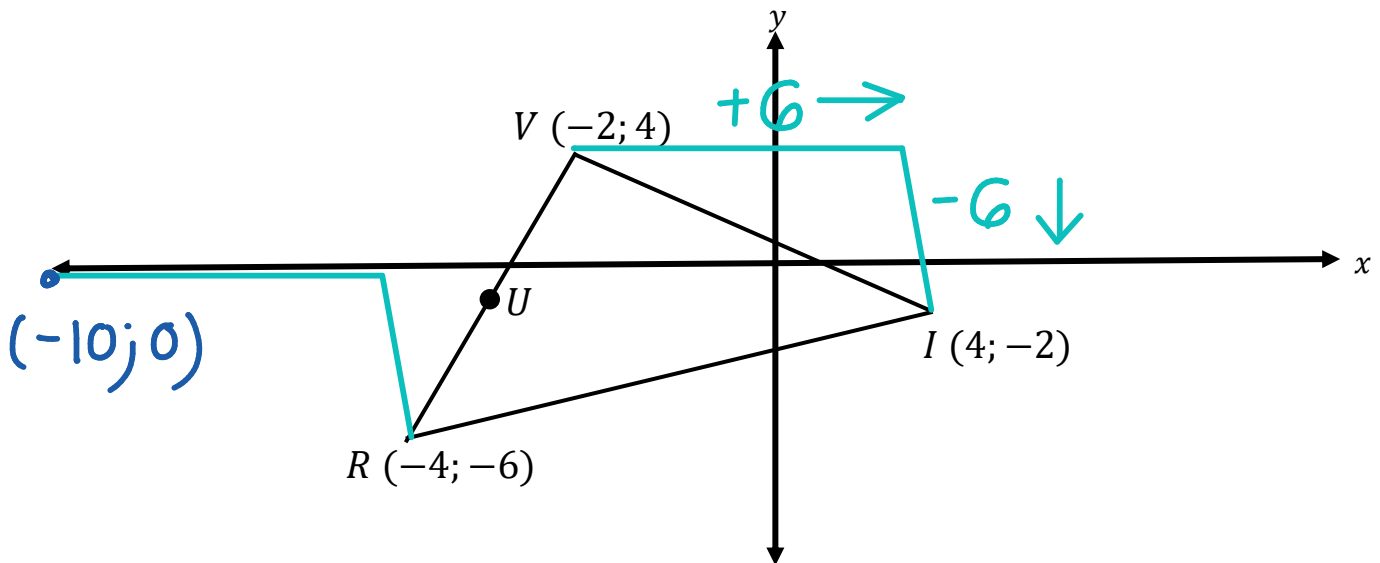
Reduce the number of infected individuals and reduce the cumulative frequency. ✓



QUESTION 2

14 MARKS

In the figure below the coordinates of the vertices of $\triangle VIR$ are $V(-2; 4)$, $I(4; -2)$ and $R(-4; -6)$



- a) Determine the length of VI , leaving your answer in simplest surd form. (2)

$$VI = \sqrt{(-2-4)^2 + (4-(-2))^2} \quad \checkmark$$

$$VI = 6\sqrt{2} \quad \checkmark$$

- b) Determine the co-ordinates of U , the mid-point of VR . (2)

$$U(-3; -1) \quad \checkmark$$

- c) Determine the gradient of VR . (2)

$$m_{VR} = \frac{4-(-6)}{-2-(-4)} \quad \checkmark$$

$$= 5 \quad \checkmark$$

d) Determine the equation of the perpendicular bisector of VR .

(3)

$$y = mx + c$$

$$y = -\frac{1}{5}x + c \quad \text{Subs } (-3; -1)$$

$$-1 = -\frac{1}{5}(-3) + c$$

$$-\frac{8}{5} = c$$

$$y = -\frac{1}{5}x - \frac{8}{5}$$

e) Determine the co-ordinates of S , given that $VIRS$ is a parallelogram.

(2)

$$S(-10; 0)$$

f) The value of q , given that $B(-1; q)$ is collinear with VI .

(3)

$$m_{VI} = \frac{4 - -2}{-2 - 4}$$

$$= -1$$

$$m_{VB} = -1$$

$$\frac{4 - q}{-2 - -1} = \frac{-1}{1}$$

$$4 - q = -1$$

$$3 = q$$

QUESTION 3

18 MARKS

a) Simplify the following, without using a calculator:

1)
$$\frac{\sin(360^\circ - \theta) - \cos(90^\circ + \theta) + \cos^2(180^\circ - \theta)}{(1 + \sin \theta)(1 - \sin \theta)} \quad (5)$$

$$= \frac{-\sin \theta + \sin \theta + \cos^2 \theta}{1 - \sin^2 \theta}$$

$$= \frac{\cos^2 \theta}{\cos^2 \theta}$$

$$= 1$$

2)
$$\frac{\sin 210^\circ - \tan 225^\circ}{\cos 180^\circ + \cos 300^\circ} - \frac{2}{\sin 510^\circ} \quad (7)$$

$$= \frac{\sin(180 + 30) - \tan(180 + 45)}{\cos 180 + \cos(360 - 60)} - \frac{2}{\sin(180 - 30)}$$

$$= \frac{-\sin 30 - \tan 45}{\cos 180 + \cos 60} - \frac{2}{\sin 30}$$

$$= \frac{-\frac{1}{2} - 1}{-1 + \frac{1}{2}} - \frac{2}{\frac{1}{2}}$$

$$= \frac{3 - 4}{-1} = -1$$

- b) If $\sqrt{3} \tan \theta = -1$ and $180^\circ < \theta < 360^\circ$, without using a calculator and with the aid of a diagram, determine the value of: $26 \sin \theta + 4\sqrt{3} \cos \theta$

1) $26 \sin \theta + 4\sqrt{3} \cos \theta$

(4)

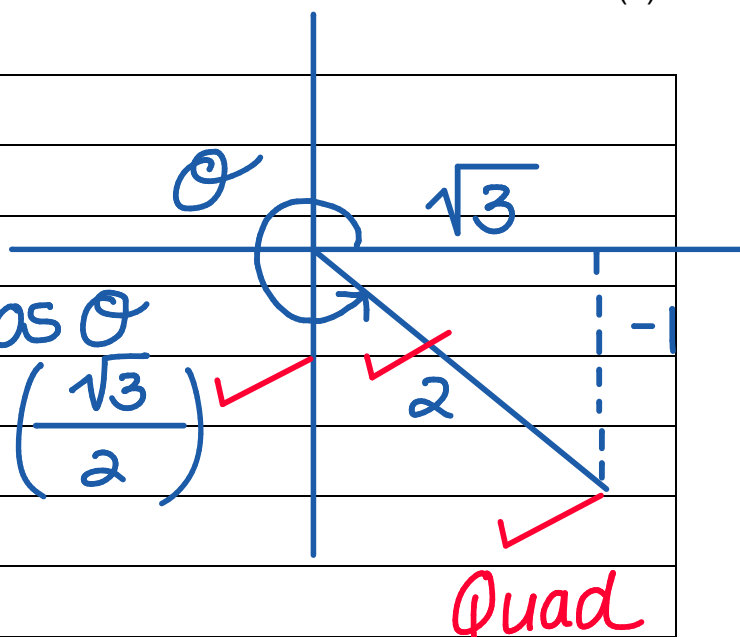
$$\tan \theta = \frac{-1}{\sqrt{3}}$$

$$26 \sin \theta + 4\sqrt{3} \cos \theta$$

$$= 26 \left(\frac{-1}{2} \right) + 4\sqrt{3} \left(\frac{\sqrt{3}}{2} \right)$$

$$= -13 + 6$$

$$= -7$$



2) θ

(2)

$$RA = 30^\circ$$

$$\theta = 360 - 30$$

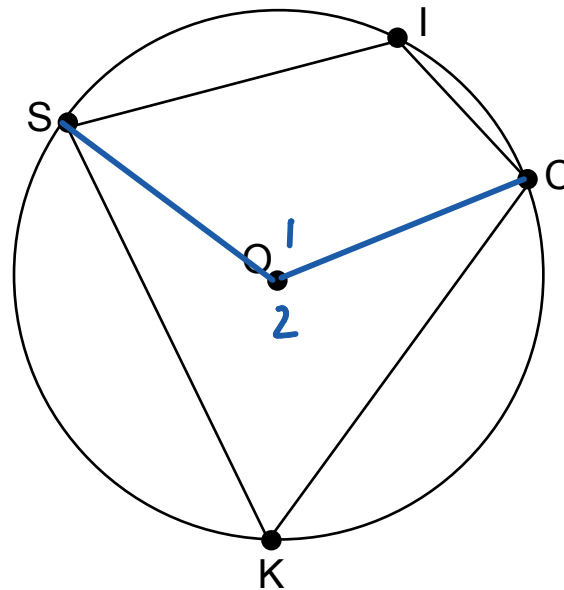
$$\theta = 330^\circ$$

QUESTION 4

14 MARKS

- a) Use the diagram below to prove the theorem that states that the opposite angles of a cyclic quadrilateral are supplementary.

(5)



Join SO and OC and label $\hat{O}_1$ and $\hat{O}_2$

Let $\hat{O}_1 = 2x$

$\hat{K} = x$ ✓ $\angle$ @ centre = $2 \times \angle$ circum

Let $\hat{O}_2 = 2y$

$\hat{I} = y$ ✓ $\angle$ @ centre = $2 \times \angle$ circum

$2x + 2y = 360$ full rev ✓

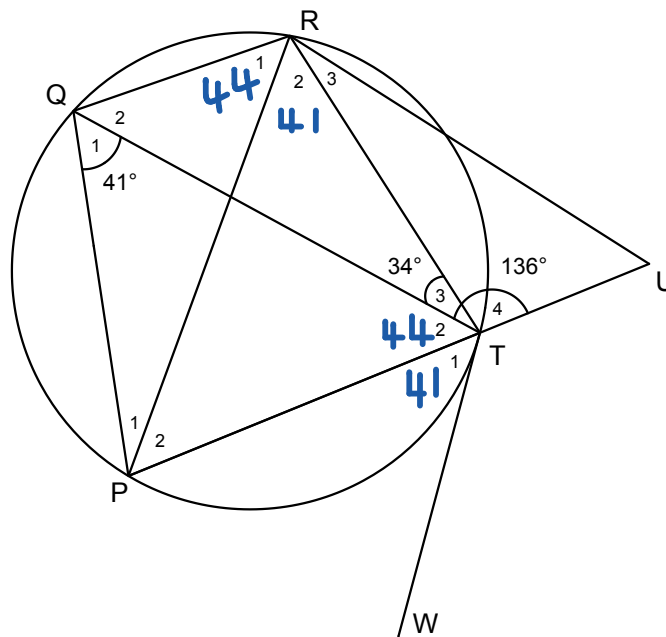
$2(x + y) = 360$

$x + y = 180^\circ$ ✓

$\therefore \hat{I} + \hat{K} = 180^\circ$

$\therefore$ opposite $\angle$'s cyclic supp.

- b) In the diagram below, points P, Q, R and T lie on the circumference of a circle. TW is a tangent to the circle at T . PT is produced to meet RU at U . $\angle Q\hat{T}U = 136^\circ$, $\angle \hat{T}_3 = 34^\circ$ and $\angle \hat{Q}_1 = 41^\circ$.



- 1) Determine the value of the following, giving reasons for your answers:

- i) $\hat{R}_2$ (2)

$\hat{R}_2 = 41^\circ$ ✓ Δ 's same seg ✓

- ii) $\hat{T}_1$ (2)

$\hat{T}_1 = 41^\circ$ ✓ tan chord thm ✓

- iii) $\hat{R}_1$ (3)

$\hat{T}_2 = 44^\circ$ Δ 's str line ✓
 $\hat{R}_1 = 44^\circ$ ✓ Δ 's same seg ✓

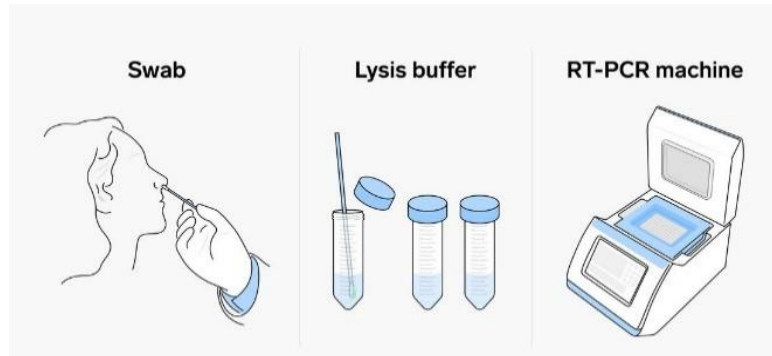
- 2) Is QT a diameter to circle $PQRT$? Justify your answer. (2)

$\hat{R}_1 + \hat{R}_2 = 85^\circ$ ✓
 $\therefore QT$ not diameter; not converse Δ 's semi circle ✓

QUESTION 5

9 MARKS

COVID-19 is diagnosed by a laboratory test called a polymerase chain reaction (PCR) molecular test. The test investigates a sample taken from a patient's respiratory tract i.e. nose, throat or chest, and then placed in a test tube and sent for testing.

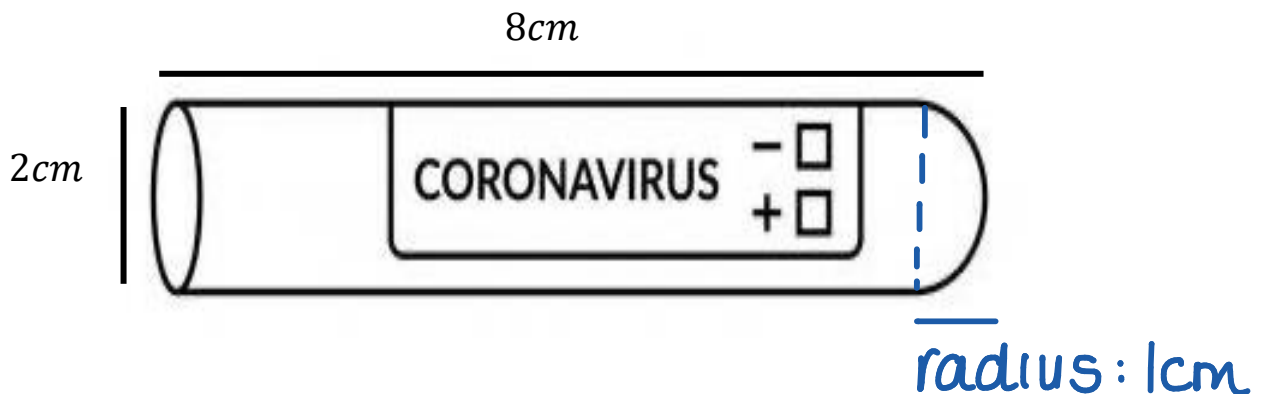


FORMULAE:

$$SA \text{ of sphere} = 4\pi r^2$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

Below is a diagram showing the test tube used to store the swab. It is a cylinder with a hemispherical end.



a) Determine the volume of the test tube.

(4)

$$\begin{aligned} \text{Volume hemisphere} &= \frac{4}{3}\pi r^3 \\ &= \frac{4}{3}\pi (1)^3 \div 2 \\ &= \frac{2}{3}\pi \\ &= 2,09\text{cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Volume cylinder} &= A \times l \\ &= \pi r^2 \times l \\ &= \pi (1)^2 \times 7 \\ &= 21,99\text{cm}^3 \end{aligned}$$

- b) It was stated that 1 069 456 tests were conducted in South Africa from 1 March to 13 June 2020. Determine the **total** amount of plastic used for these tests. (5)

$$SA_{\text{cylinder}} = 2\pi r \times h$$

$$= 2\pi(1) \times 7$$

$$= 14\pi$$

$$= 43,98\text{cm}^2 \checkmark$$

$$SA_{\text{hemisphere}} = 4\pi r^2$$

$$= 4\pi(1)^2 \div 2 \checkmark$$

$$= 2\pi$$

$$= 6,28\text{cm}^2 \checkmark$$

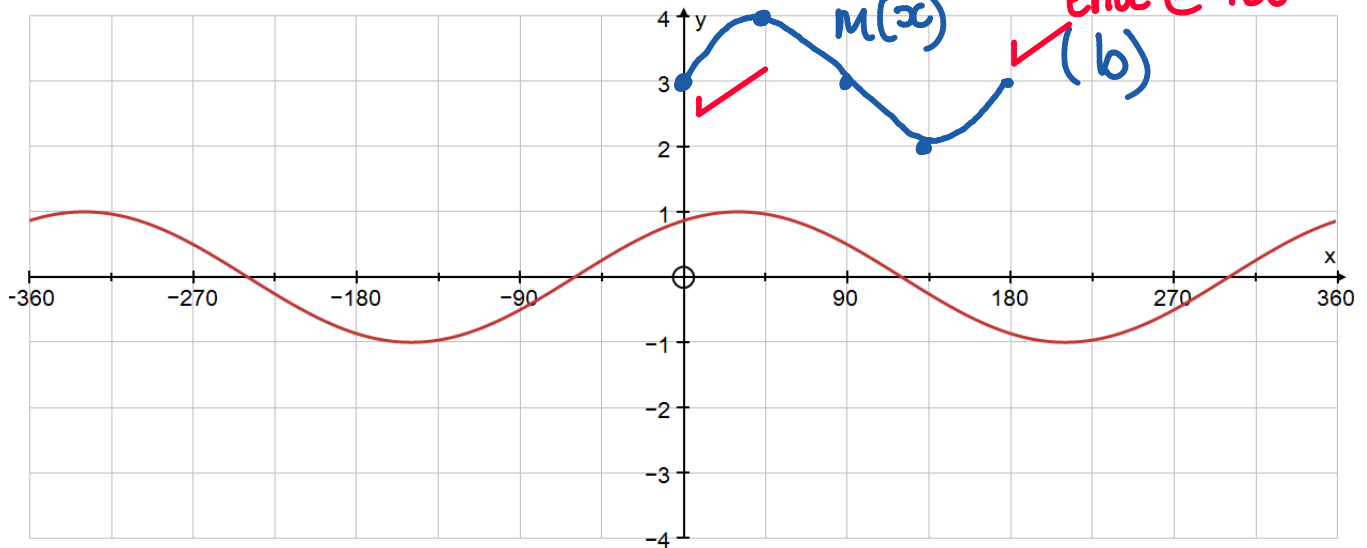
$$\text{Total surface area} = 16\pi \times 1069\,456 \checkmark$$

$$= 53\,756\,721,81\text{cm}^2 \checkmark$$

QUESTION 6

12 MARKS

Given the function: $f(x) = \cos(x - 30^\circ)$ for $x \in [-360^\circ; 360^\circ]$



a) Determine:

1) the period of the function f

(1)

360° ✓

2) the period of the function g , if $g(x) = f(2x)$

(2)

$$f(2x) = \cos(2x - 30)$$

period: 180° ✓✓

3) the range of the function h , if $h(x) = f(x) - 1$

(2)

$y \in [-2; 0]$ ✓✓

4) the maximum of function q , if $q(x) = \frac{1}{2}f(x) + 2$

(2)

$$q(x) = \frac{1}{2}\cos(x - 30) + 2$$

maximum: $2\frac{1}{2}$ ✓✓

5) the equation of k , in simplified form, if k is obtained by shifting f 60° to the right. (2)

$$K(x) = \cos(x - 60 - 30)$$

$$K(x) = \cos(x - 90)$$

$$K(x) = \cos(270 + x)$$

$$K(x) = \sin x$$
 ✓

b) Sketch $m(x) = \sin(2x) + 3$ on the axes on page 14 for $x \in [0^\circ; 180^\circ]$

(3)

$$\begin{array}{ll} \text{min: } -1 + 3 = 2 & \text{step: } \frac{180}{4} \\ \text{max: } 1 + 3 = 4 & = 45 \end{array}$$

END OF SECTION A

SECTION B:

QUESTION 7

12 MARKS

The table below summarizes the total tests conducted compared to the total deaths in 6 countries across the globe. The data was obtained on 1 October 2020.

#	Country	Total Tests (x)	Total Deaths (y)
1	USA	107 538 839	211 752
10	South Africa	4 187 917	16 734
32	Netherlands	2 449 577	6 406
56	Singapore	2 885 545	27
71	Ireland	1 180 984	1 804
123	Hong Kong	3 311 279	105

- a) Determine the mean and standard deviation of the total deaths from the above data. (2)

$$\bar{y} = 39\,471,33 \quad \sigma_y = 77\,260,73$$

- b) What percentage of the above listed countries fall within one standard deviation of the mean number of deaths? (2)

$$(-37\,789,4; 116\,732,06)$$

$$\frac{5}{6} \times 100 = 83,33\%$$

- c) Calculate the equation of the least squares regression line. Rounding your answer to five decimal digits. (2)

$$y = -536,09269 + 0,00197x$$

- d) Lesotho conducted 18 610 tests. Using your line of best fit, predict the number of deaths they have had, and comment on the validity of your result justifying your answer. (3)

$$y = -499,34$$

Very unreliable, extrapolation

- e) Calculate the correlation coefficient for the above data and then comment on the strength of the relationship, rounding your answer to 3 decimal digits. (3)

$$0,998$$

Very strong (almost perfect), positive.

QUESTION 8

18 MARKS

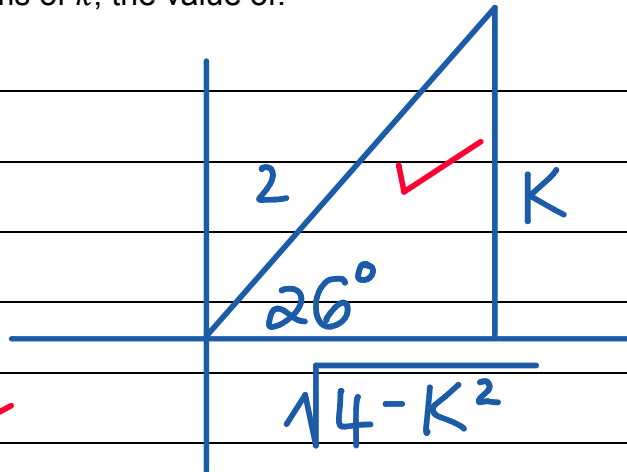
a) Given: $\sin 26^\circ = \frac{k}{2}$. Determine, in terms of k , the value of:

1) $\cos 64^\circ$

(3)

$$= \sin 26^\circ$$

$$= \frac{k}{2}$$



2) $\cos 26^\circ$

(2)

$$= \frac{\sqrt{4 - k^2}}{2}$$

3) $\tan 64^\circ$

(2)

$$= \frac{\sin 64^\circ}{\cos 64^\circ}$$

$$= \frac{\sqrt{4 - k^2}}{2} \times \frac{2}{k}$$

$$= \frac{\cos 26^\circ}{\sin 26^\circ}$$

$$= \frac{\sqrt{4 - k^2}}{k}$$

b) Prove the following identity: $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = \frac{2}{\sin \theta}$

(4)

$$\text{LHS} = \frac{(1 + \cos \theta)(1 + \cos \theta) + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2 \cos \theta + 2}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2 (\cos \theta + 1)}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2}{\sin \theta}$$

c) Solve for x : $\sin(x + 30^\circ) = -0,2$ where $x \in (-180^\circ; 0)$.

(3)

$$CA = -11,54$$

$$x + 30 = -11,54 + K \cdot 360 \quad x + 30 = 180 - (-11,54) + K \cdot 360$$

$$x = -41,54 + K \cdot 360 \quad x = 161,54 + K \cdot 360$$

$$\underline{x \in \{-41,54\}}$$

d) $\cos 2x$; $\sin 2x$; $\sqrt{2} \sin 2x$ are the first 3 terms of an arithmetic number pattern
($T_2 - T_1 = T_3 - T_2$). Determine the general solution for x .

(4)

$$\sin 2x - \cos 2x = \sqrt{2} \sin 2x - \sin 2x$$

$$-\cos 2x = (-2 + \sqrt{2}) \sin 2x$$

$$-1 = (-2 + \sqrt{2}) \tan 2x$$

$$\underline{\frac{-1}{-2 + \sqrt{2}}} = \tan 2x$$

$$CA = 59,64$$

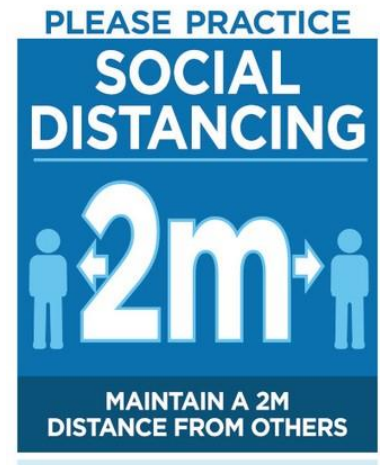
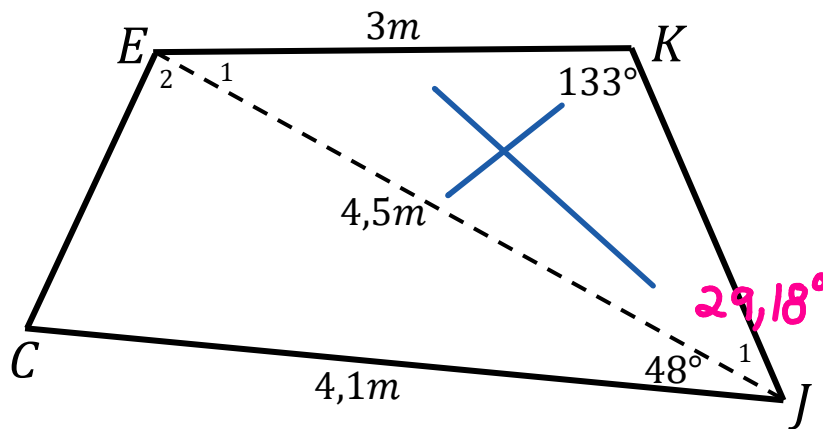
$$2x = 59,64 + K \cdot 180$$

$$\underline{x = 29,82 + K \cdot 90^\circ}$$

QUESTION 9

11 MARKS

The virus that causes COVID-19 is mainly transmitted through droplets generated when an infected person coughs, sneezes, or exhales. By maintaining a $2m$ space between people it reduces the spread of the infection. Eli (E), Kai (K), James (J) and Cody (C) are four friends standing together at break.



- a) Determine the size of $\hat{J}_1$ (3)

$$\frac{\sin 133}{4,5} = \frac{\sin \hat{J}_1}{3}$$

$$\sin \hat{J}_1 = 0,48757$$

$$\hat{J}_1 = 29,18^\circ$$

- b) Determine the length of KJ and hence determine if Kai and James are practising correct social distancing. (4)

$$\hat{E}_1 = 17,82$$

$$\frac{\sin 17,82}{KJ} = \frac{\sin 133}{4,5}$$

$$KJ = 1,88m$$

$\therefore$ No they are not practising correct social distancing.

c) Determine the area between the 4 boys.

(4)

$$\begin{aligned} \text{Area } \triangle EKT &= \frac{1}{2} (3)(4,5) \cdot \sin 17,82 \\ &= 2,07 \text{ m}^2 \end{aligned}$$

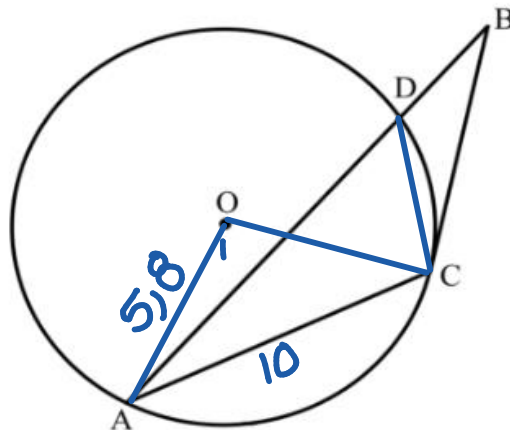
$$\begin{aligned} \text{Area } \triangle ECT &= \frac{1}{2} (4,5)(4,1) \cdot \sin 48 \\ &= 6,86 \text{ m}^2 \end{aligned}$$

$$\therefore \text{Total area} = 8,93 \text{ m}^2$$

QUESTION 10

5 MARKS

O is the centre of the circle passing through A, C and D. Chord AD is produced to B and BC is a tangent to the circle at C. $AC = 10\text{cm}$ and the radius of the circle is $5,8\text{cm}$.



Calculate the size of $\hat{ADC}$ rounded off to the nearest degree. (HINT: Draw in AO, OC and DC)

Construct AO, OC and DC

$$\begin{aligned} 10^2 &= 5,8^2 + 5,8^2 - 2(5,8)(5,8) \cdot \cos \hat{O}_1 \\ -409 &= \cos \hat{O}_1 \\ 841 \end{aligned}$$

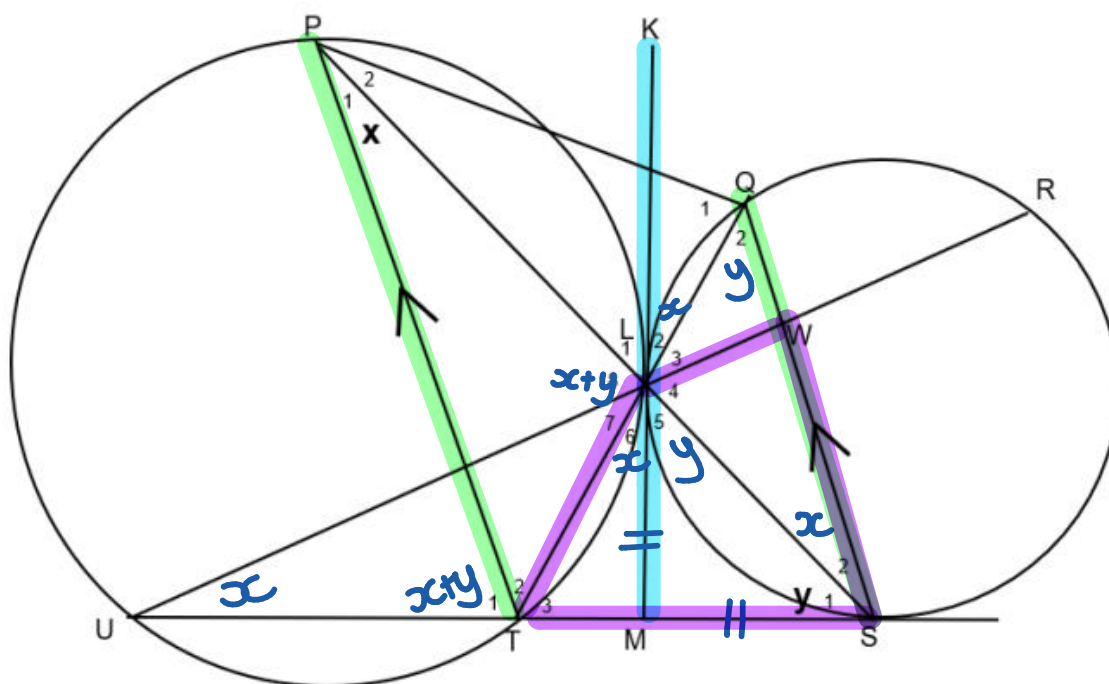
$$119,1 = \hat{O}_1$$

$$\hat{ADC} = 59,55^\circ \quad \angle @ \text{ centre} = 2 \times \angle \text{circum.}$$

QUESTION 11

12 MARKS

In the diagram below, two circles touch each other externally at point L . KLM is a common tangent to both circles at L . $UTMS$ is a tangent to circle $QRSL$ at S . $PT \parallel QS$. Let $\hat{P}_1 = x$ and $\hat{S}_1 = y$.



a) Name with reasons, 4 other angle equal to x

(4)

$\hat{S}_2 = x$	alt $\angle$'s; $PT \parallel QS$ ✓
$\hat{L}_6 = x$	tan chord thm. ✓
$\hat{L}_2 = x$	vert opp ✓
$\hat{U} = x$	$\angle$'s same seg ✓

b) Name with reasons, THREE other angles equal to y

(3)

$\hat{Q}_2 = y$	tan chord thm ✓
$\hat{L}_5 = y$	isos Δ (equal tan) ✓
$\hat{L}_1 = y$	vert opp ✓

c) Show that $\hat{L}_8 = x + y$

(2)

$$\hat{T}_1 = x + y$$

$$\hat{L}_8 = x + y$$

Corresp Δ 's; PT || QS ✓

Δ 's same seg. ✓

d) Determine the numerical value of $x + y$ if it is given that $TSWL$ is a cyclic quadrilateral. (3)

$$\hat{L}_4 = x + y$$

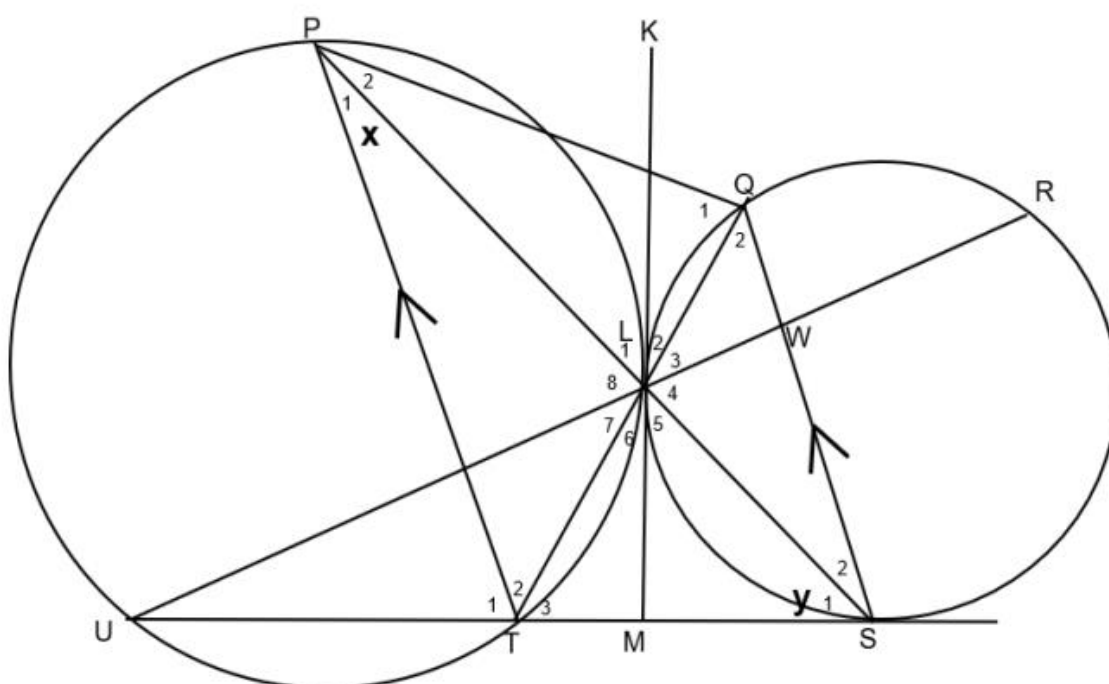
vert opp ✓

$$3(x + y) = 180$$

$$x + y = 60^\circ$$

opp Δ 's cyclic ✓

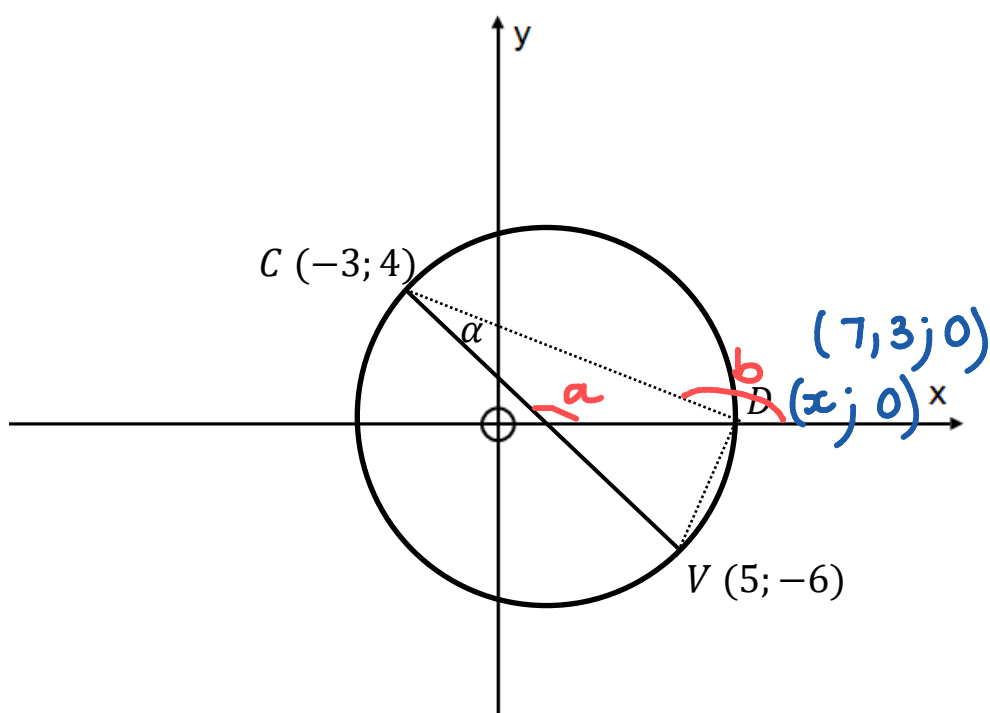
REPEATED DIAGRAM:



QUESTION 12

14 MARKS

CV is the diameter of circle CVD.



- a) Determine the length of the radius of the circle leaving your answer in simplest surd form. (3)

$$CV = \sqrt{(-3-5)^2 + (4-(-6))^2} \quad \checkmark$$

$$CV = 2\sqrt{41} \quad \checkmark$$

$$\therefore \text{Radius} = \sqrt{41} \quad \checkmark$$

- b) Why is $\angle C\hat{D}V = 90^\circ$? (1)

Δ 's semi circle $\checkmark$

- c) Hence, or otherwise, show that the co-ordinates of D is (7,3; 0), rounded to one decimal place, given that D is a point on the x -axis. (5)

$$D(x;0)$$

$$m_1 \times m_2 = -1$$

$$\frac{4-0}{-3-x} \times \frac{0--6}{x-5} = -1$$

$$\frac{24}{(-3-x)(x-5)} = \frac{-1}{1}$$

$$-24 = -3x + 15 - x^2 + 5x$$

$$0 = -x^2 + 2x + 39$$

$$x = 1 + 2\sqrt{10} \quad \vee \quad x = 1 - 2\sqrt{10}$$

$$\underline{x = 7,3} \quad \vee \quad x = -5,3$$

- d) Determine the magnitude of α . (5)

$$\tan a = m$$

$$\tan a = \frac{4--6}{-3-5} \quad \vee$$

$$a = 128,66 \quad \vee$$

$$\underline{\alpha = 30,12^\circ}$$

$$\tan b = m$$

$$\tan b = \frac{4-0}{-3-7,3} \quad \vee$$

$$b = -21,22 + 180$$

$$b = 158,78 \quad \vee$$

$$\text{ext } \angle \text{'s } \triangle \quad \vee$$

ADDITIONAL SPACE (ALL questions)

**REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE ALL ANSWERS ARE MARKED.**

[illegible]

